Medical Device Recall Risk Analytics Using R

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Abstract

This project applies survival analysis and business analytics techniques to medical device recall data to identify factors associated with recall resolution times and support risk - based monitoring and resource allocation. The analysis included 404 medical device recalls and examined several predictors, including device classification, medical specialty, root cause description, number of products involved, number of associated 510(k) numbers which represent the count of FDA premarket notification identifiers recorded for each medical device recall, and recall reason description length. Kaplan - Meier estimation was used to characterize recall resolution patterns, while Cox proportional hazards regression was initially employed to evaluate associations between recall characteristics and resolution hazards. However, diagnostic testing revealed violations of the proportional hazards assumption, motivating the use of parametric accelerated failure time (AFT) models. Four candidate survival distributions were compared using the Akaike Information Criterion (AIC), with log - logistic distribution providing the best relative fit. Natural cubic splines were subsequently incorporated to accommodate potentially nonlinear relationships between numerical predictors and recall resolution times. The spline enhanced log - logistic AFT model demonstrated a significant improvement over the original linear effects model ($\chi^2 = 64.454, df = 6, p - value < 0.001$) with an AIC reduction of 52.454. Device classification, medical specialty, and several recall root causes were significantly associated with resolution duration. A Power BI dashboard was developed to translate analytical findings into accessible business insights for recall monitoring and decision making. This project demonstrates how survival analysis and business intelligence tools can be integrated to support data driven medical device recall risk assessment, resource prioritization, and extensions into other fields.

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1 INTRODUCTION

The medical device recall data is a synthetic dataset obtained from Kaggle.com and consists of 407 medical device observations each with 32 characteristics. For this analysis, I retained 8 variables relevant to risk analytics including follow - up time (in days), event status, device classification, medical specialty, root cause description, number of products involved in the recall, number of associated 501k numbers, and reason for recall. Each variable was converted to its appropriate data type, with categorical variables assigned as factors and numerical variables assigned as numeric. Observations containing missing values were removed to ensure a complete dataset for statistical modeling. To address sparse categorical levels and improve model stability, root cause descriptions and medical specialties with fewer than 10 observations were grouped into an "Other" category. Similarly, device classes 3 and N were combined into a single category due to their limited representation. The resulting dataset for analysis contained 404 observations and eight variables, providing a structured dataset for fitting Cox proportional hazards and accelerated failure time (AFT) models to examine factors associated with the time to medical device recall event resolution.

1.1 Descriptive Statistics

Frequency and relative frequency analyses were conducted to examine the distribution of categorical variables across the 404 medical device recall observations. Of these observations, 359 (88.9%) experience the event of interest, while 45 (11.1%) were censored. Regarding device classification, Class II devices accounted for the majority of observations (81.9%), followed by Class I devices (13.6%) and the Other/Rare category (4.46%).

Medical specialty distributions indicated that General Hospital devices represented the largest proportion of observations (28.2%), followed by Orthopedic (11.4%), and Neurology (10.1%).

Regarding recall root causes, Device Design was the most frequented reported category (24.5%) followed by Nonconforming Material/Component (16.8%), and Process Control (11.9%).

Overall, the descriptive analysis revealed substantial differences in category proportions, particularly the high number of Class II devices and design - related recall causes. These findings provide an overview of the dataset and highlight potential imbalances among categorical predictors that should be considered when interpreting the subsequent Cox proportional hazards and accelerated failure time (AFT) models.

Table 1: Frequency and Relative Frequency of Event Observed

event_observed	Frequency	Relative_Frequency	Percent
0	45	0.111	11.139
1	359	0.889	88.861

Table 2: Frequency and Relative Frequency of Device Class

device_class	Frequency	Relative_Frequency	Percent
1	55	0.136	13.614
2	331	0.819	81.931
Other/Rare	18	0.045	4.455

Table 3: Frequency and Relative Frequency of Medical Specialty

medical_specialty	Frequency	Relative_Frequency	Percent
Anesthesiology	11	0.027	2.723
Cardiovascular	27	0.067	6.683
Clinical Chemistry	21	0.052	5.198
Dental	10	0.025	2.475
Gastroenterology, Urology	10	0.025	2.475
General Hospital	114	0.282	28.218
General, Plastic Surgery	38	0.094	9.406
Neurology	41	0.101	10.149
Orthopedic	46	0.114	11.386
Radiology	28	0.069	6.931
Unknown	17	0.042	4.208
Other/Rare	41	0.101	10.149

Table 4: Frequency and Relative Frequency of Root Cause Description

root_cause_description	Frequency	Relative_Frequency	Percent
Device Design	99	0.245	24.505
Nonconforming Material/Component	68	0.168	16.832
Process change control	35	0.087	8.663
Process control	48	0.119	11.881
Software design	20	0.050	4.950
Under Investigation by firm	45	0.111	11.139
Unknown/Undetermined by firm	18	0.045	4.455
Use error	12	0.030	2.970
Other	59	0.146	14.604

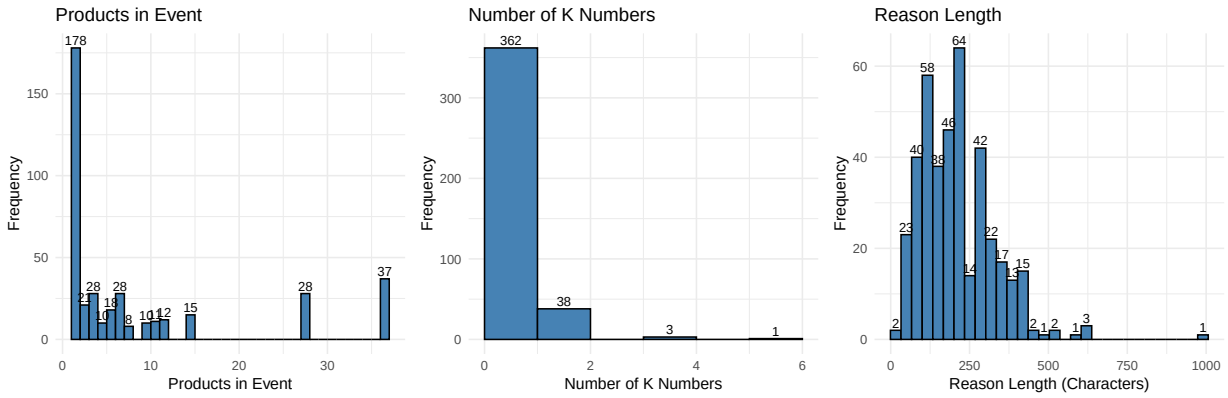


Figure 1: Descriptive Statistics: Histograms for Products in Event, Number of K Numbers, and Reason Length.

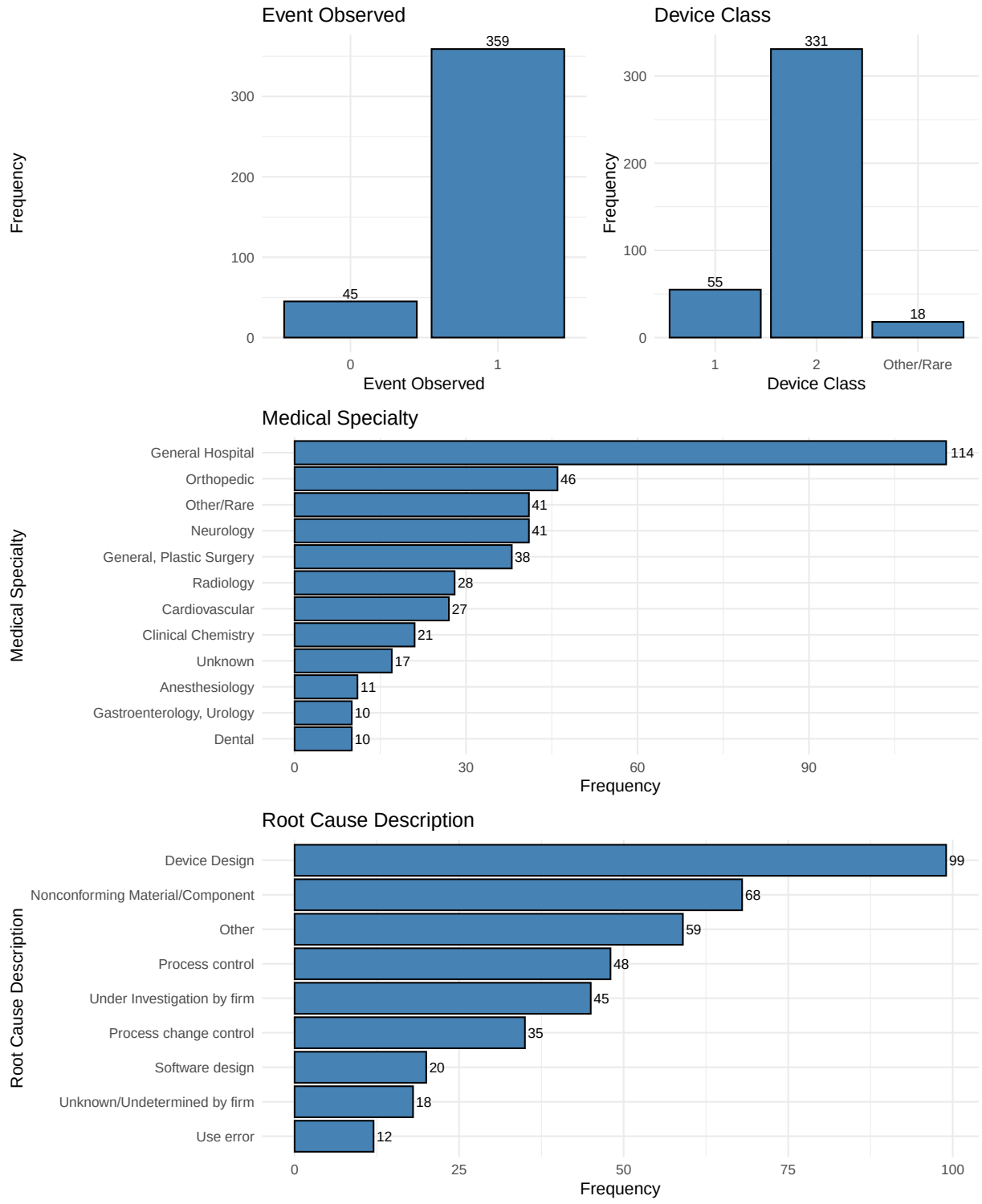


Figure 2: Frequency Distributions for Medical Device Recalls: Event Observed, Device Class, Medical Specialty, and Root Cause Description.

Two Kaplan - Meier survival analyses were conducted to estimate the probability of medical device recalls remaining unresolved over time. The analyses included the 404 recall observations, of which 359 (88.9%) experienced recall resolution and 45 (11.1%) were censored.

In our first Kaplan Meier analysis, the survival curve demonstrated a progressive decline in the probability of remaining unresolved as the duration since recall initiation increased. The median time to recall resolution is 692 days, indicating that half of recalls had been resolved within this time period. At 1000 days, the estimated probability of recalls remaining unresolved decreased to approximately 25%, followed by approximately 15% at 2,000 days. Beyond 2,500 days, the survival probability approached roughly 5% suggesting that a relatively small proportion of recalls remained unresolved over extended periods. The shaded area represents a confidence interval and it illustrates uncertainty surrounding the survival estimates, while the censoring markers identify observations for which the exact resolution time was not observed.

In our second survival analysis, Kaplan -Meier curves were constructed to compare recall resolution times across device classifications. The analysis included 55 Class I devices (13.6%), 331 Class II devices (81.9%), and 18 (4.46%) grouped under the Other/Rare category. All three devices classifications demonstrated a progressive decline in the probability of remaining unresolved over time, with substantial overlap among the survival curves. Class I devices generally exhibited slightly lower survival probabilities during portions of the follow - up period, suggesting potentially shorter recall resolution times compared with Class II devices. However, these differences were relatively small, and the confidence intervals overlapped considerably. The Other/Rare classification exhibited greater uncertainty, as reflected by its wider confidence interval, primarily due to its smaller sample size. A log rank test was conducted to assess differences in survival distributions across device classifications, yielding a p - value of 0.56. Since this p - value exceeds the conventional significance threshold of 0.05, the null hypothesis of equal survival distributions was not rejected. These findings indicate insufficient statistical evidence of differences in recall resolution times among device classifications.

Overall, these Kaplan - Meier analyses provide a nonparametric assessment of recall resolution times and serve as baselines for the subsequent Cox proportional hazards and accelerated failure time modeling.

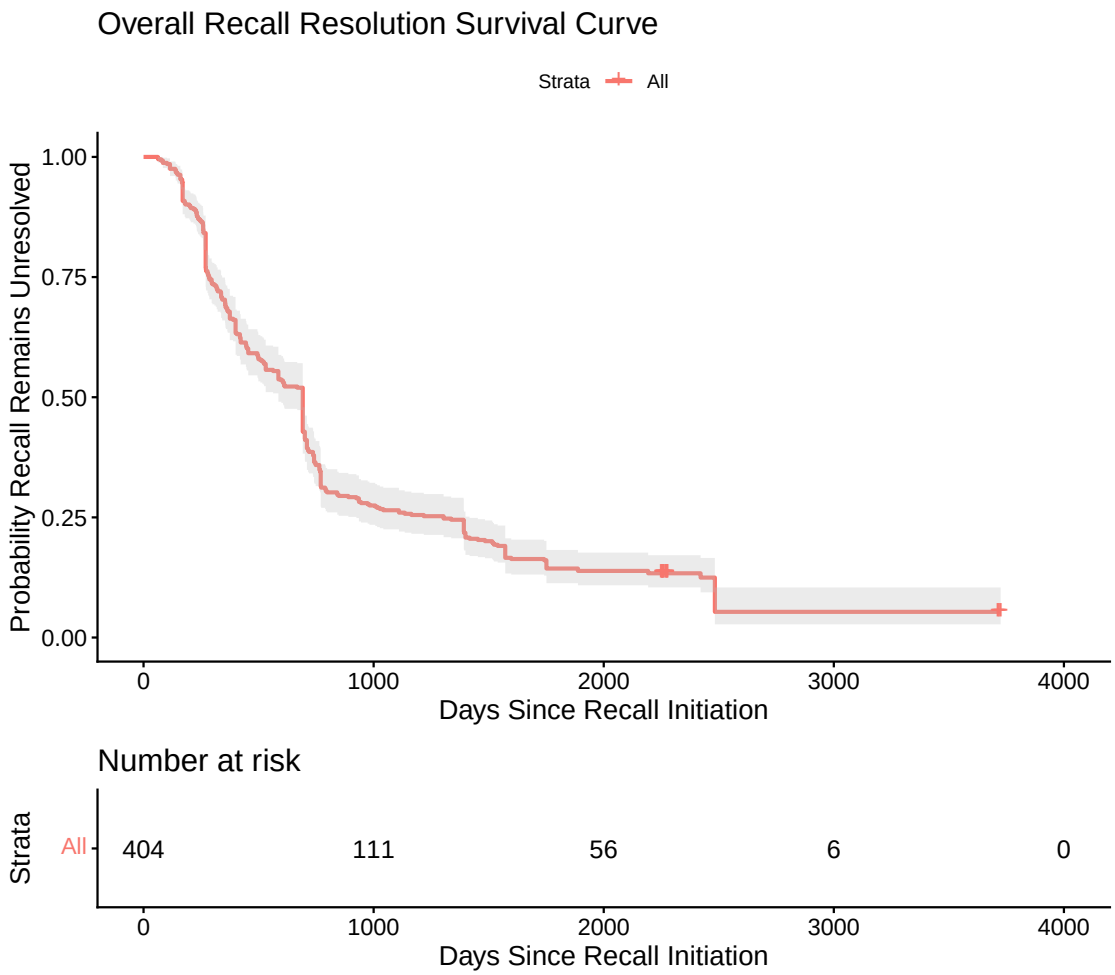


Figure 3: Kaplan-Meier Survival Curve with Risk Table.

The stratified log-rank assessment and corresponding survival profiles highlight the variation in resolution rates across distinct medical device classes.

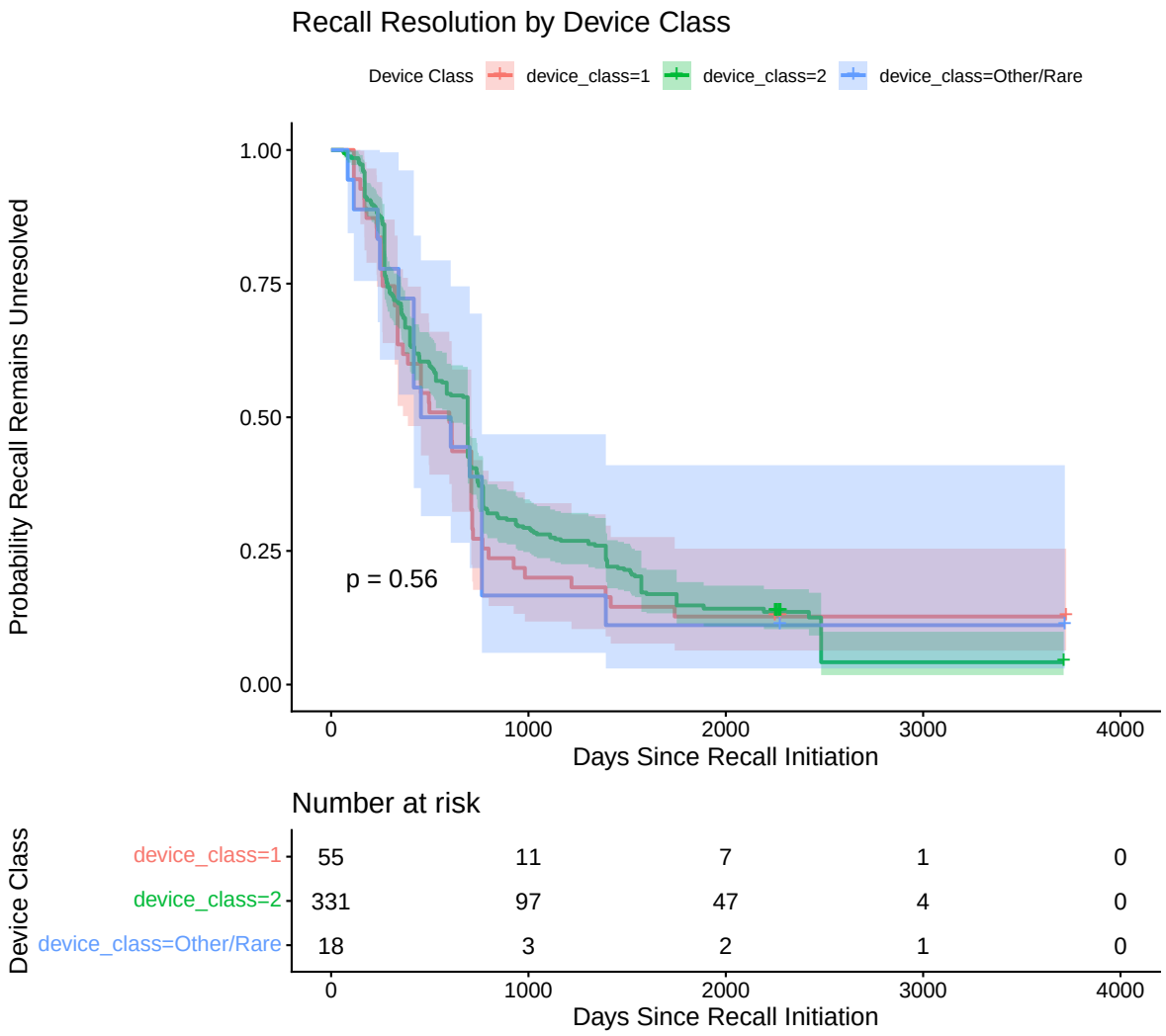


Figure 4: Kaplan-Meier Curve Grouped by Device Class with Log-Rank p-value and Number at Risk Table.

1.2 Data Preparation

Following the data preprocessing and exploratory analysis, the medical device recall dataset was randomly partitioned into training and testing subsets using an 80/20 split. From the 404 available observations, 323 (80%) were allocated to the training dataset, while the remaining 81 (20%) were reserved for testing. A random seed of 1812 was specified to ensure reproducibility of the data partitioning process. The training dataset was used to fit and estimate the parameters of the Cox proportional hazards and accelerated failure time models, while the testing dataset was reserved for evaluating predictive performance. Additionally, unused factor levels were removed from both subsets to eliminate categories that were no longer represented

following the random split. This partitioning approach allows for an independent assessment of model performance and helps evaluate the ability of the fitted survival models to generalize beyond the training data.

5).

Table 5: Distribution and Proportion of Events Across Training and Testing Subsets

	Training Subset (n_{train})		**Testing Subset (n_{test})**	
Event Observed Status	**Count (n)**	**Proportion (%)**	**Count (n)**	**Proportion (%)**
No Event Observed (0)	85	29.82	37	30.33
Event Observed (1)	200	70.18	85	69.67
Total Observations	**285**	**100.00**	**122**	**100.00**

2 SURVIVAL ANALYSIS

2.1 Cox Proportional Hazards Model

A multivariate Cox proportional hazards regression model was fitted using the training dataset to investigate factors associated with medical device recall resolution times. The model included 323 observations, of which 287 experienced recall resolution, and it incorporated the following variables: device classification, medical root specialty, root cause description, number of products involved, number of associated 510(k) numbers, and recall reason description length as predictors.

$$h(t \mid X) = h_0(t) \exp(X\beta) \tag{1}$$

$$h(t \mid X) = h_0(t) \exp\Big(- 0.9112(\text{device_class2}) + \cdots + 0.0000994(\text{reason_length_chars})\Big) \tag{2}$$

The general Cox model is shown in Equation 1, and the fitted model is shown in Equation 2.

The Cox model demonstrated statistical significance with the likelihood ratio test, with test statistic, degrees of freedom, and p - value of $\chi^2 = 92.82$, df = 24, p - value ¡ 0.001 respectively. The Wald test (χ^2 = 94.66, df = 24, p - value ¡ 0.001) and score test ($\chi^2 = 106.50$, df = 24, p - value ¡ 0.001) further support statistical significance. The model achieved a concordance index of 0.657, suggesting moderate discriminative ability within the training dataset.

Several predictors exhibited statistically significant associations with recall resolutions hazards. Class II devices demonstrated a lower hazard of recall resolution compared with Class I devices (HR = 0.402, 95% CI: (0.251,0.643), p - value ¡ 0.001). Among medical specialties, Neurology (HR = 2.871, p - value = 0.039) and Orthopedic (HR = 2.746, p - value = 0.026) were associated with higher resolution hazards relative to anesthesiology. Compared with device design, root causes including "nonconforming material/component" (HR = 2.115, p - value = 0.003), "process control" (HR = 1.998, p - value = 0.007), "under investigation by firm" (HR = 2.043, p - value = 0.012), "unknown/undetermined by firm" (HR = 2.928, p - value = 0.004), and "use error" (HR = 2.526, p - value = 0.023) were associated with higher resolution hazards.

Among numerical predictors, the "number of products involved in a recall" (HR = 1.031, p - value ¡ 0.001) and the "number of associated 510(k) numbers" (HR = 1.442, p - value 0.005) were positively associated with recall resolution hazards. In contrast, "recall reason description length" was not statistically significant (p - value = 0.850)

2.2 Evaluation of the Proportional Hazards Assumption

The proportional hazards assumptions of the Cox regression model was assessed using scaled Schoenfeld residuals and the cox.zph() test in R. The global test produced a chi - square statistic of $\chi^2 = 189.95$ with df = 24, p - value ¡ 0.001, providing strong evidence against the proportional hazards assumption. Individual tests indicated significant violations for "medical specialty", "root cause description", "number of products involved in the recall", and "recall reason description length". (Table 6).

Table 6: Schoenfeld Residual Proportional Hazards Test Summary Matrix

Variable	Chisq (chisq)	df	p-value (p)
device_class	2.45	2	0.2936
medical_specialty	63.19	11	0.0000
root_cause_description	69.73	8	0.0000
products_in_event	65.16	1	0.0000
n_k_numbers	0.12	1	0.7260
reason_length_chars	5.55	1	0.0184
GLOBAL	189.95	24	0.0000

The localized functional deviation trajectories across the covariate metrics are visually mapped in Figure

5.

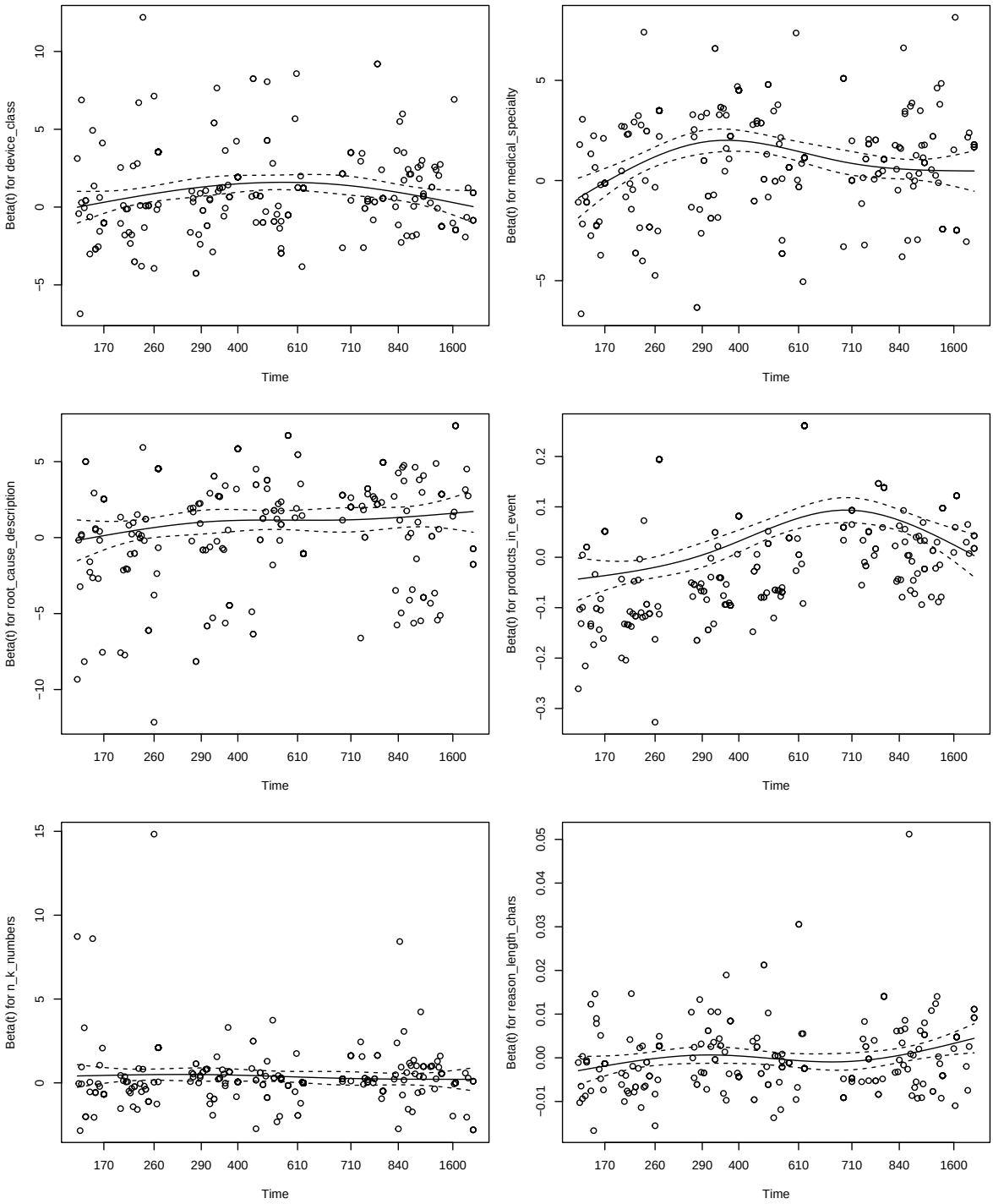


Figure 5: Scaled Schoenfeld Residual Curves for Cox Predictors over Follow-Up Duration (3×2 Grid Matrix Layout).

These findings suggest that the associations between several predictors and recall resolution hazards may change over time, violating the assumption of constant hazard ratios required by the standard Cox proportional hazards model. Consequently, the estimated hazard ratios should be interpreted cautiously, as they may not adequately represent the effects of these predictors throughout the follow - up period. Given these diagnostic results, accelerated failure time (AFT) models were considered as an alternative modeling approach because they do not require proportional hazards. However, the suitability of an AFT model is evaluated through its own distributional assumptions, model diagnostics, and goodness of fit measures.

3 PARAMETRIC ACCELERATED FAILURE TIME (AFT) ALTERNATIVES

Following the proportional hazards assumption violation identified in the Cox regression analysis, parametric accelerated failure time (AFT) models were evaluated as ana alternative approach for modeling medical device recall resolution times. Four candidate survival distributions were considered: Weibull, exponential, lognormal, and log - logistic. Each model was fitted using the same training dataset of 323 observations and included "device classification", medical specialty", "root cause description", "number of products involved in the recall", "number associated 510(k) numbers", and "recall reason description length" as predictors.

$$\log(T_i) = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ij} + \sigma\varepsilon_i \tag{3}$$

Here, T_i represents time to recall resolution, β_j are regression coefficients, X_{ij} are predictors, σ is the scale parameter, and the error ϵ_i follows a distribution determined by the selected AFT model.

3.1 AIC Optimization Ranking

Model comparison was performed using the Akaike Information Criterion (AIC), which evaluates the relative goodness of fit while penalizing model complexity. The log logistic AFT model achieved the lowest AIC (4400.023), followed by the lognormal(4416.00), Weibull(4443.53), and exponential (4476.49). The AIC difference between the log - logistic and the next best lognormal model was 16.58, indicating substantially stronger relative support for the log - logistic distribution among the candidate models. (Table 7).

Table 7: Information-Theoretic Model Selection Matrix (AIC Optimization Ranking)

Parametric Distribution (AFT Model)	Degrees of Freedom	AIC Metric
Log-Logistic	26	4400.02
Log-Normal	26	4416.60
Weibull	26	4443.53
Exponential	25	4476.49

Based on these findings, the log - logistic AFT model was selected for further analysis. Further, several predictors demonstrated statistically significant associations with recall resolution times. Regression coefficients from the log - logistic accelerated failure time (AFT) model were exponentiated to obtain time ratios (T), which quantify the multiplicative association between predictor variables and recall resolution duration. A time ratio greater than one indicates a longer predicted time to recall resolution, whereas a time ratio below one indicates a shorter predicted duration, holding other predictors constant. For example, Class II devices exhibited a time ratio of 2.15 relative to Class I devices, indicating approximately 115.1% longer predicted resolution times. Conversely, recalls attributed to "Nonconforming Material/Component" had a time ratio of 0.475 relative to "device design" recalls, corresponding to approximately 52.5% shorter predicted resolution times. The AFT framework therefore provides a direct interpretation of predictor effects in terms of recall resolution duration rather than instantaneous event hazards.

Class II devices were associated with longer predicted resolution times compared with Class I devices ($beta = 0.7659, TR = 2.151, p - value < 0.001$). Similarly, devices classified under "General/Plastic Surgery" exhibited longer predicted resolution times relative to "Anesthesiology ($beta = 0.6846, TR = 1.983, p-value < 0.034$). Conversely, "Orthopedic" devices were associated with shorted predicted resolution times ($beta = -0.7817, TR = 0.458, p - value < 0.016$).

Among recall root causes, "Nonconforming Material/Component" ($TR = 0.475, p - value < 0.001$), "Under Investigation by Firm ($TR = 0.628, p - value = 0.017$), "Unknown/Undetermined by Firm" ($TR = 0.476, p - value = 0.003$) and "Other" ($TR = 0.589, p - value = 0.007$). were significantly associated with shorter predicted resolution times relative to "device design".

For numerical predictors, the number of products involved in a recall ($beta = -0.0158, TR = 0.984, p - value = 0.008$) and the "number of associated 510(k) numbers (($beta = -0.4069, TR = 0.666, p - value < 0.001$) were associated with shorter predicted resolution times. In contrast, "recall reason description length"

was not statistically significantly (($p - value < 0.857$).

Overall, the log - logistic AFT model identified several statistically significant associations between medical device characteristics and recall resolution durations. These findings provide an alternative time based interpretation to the Cox proportional hazards model, for which the proportional hazards assumption was violated.

$$\begin{aligned}
\log(T) = 6.8741 &+ 0.7659(\text{Device Class 2}) - 0.6421(\text{Other/Rare Device Class}) \\
&- 0.3384(\text{Cardiovascular}) - 0.0560(\text{Clinical Chemistry}) - 0.4185(\text{Dental}) \\
&- 0.6323(\text{Gastroenterology/Urology}) - 0.1124(\text{General Hospital}) \\
&+ 0.6846(\text{General/Plastic Surgery}) - 0.5339(\text{Neurology}) \\
&- 0.7817(\text{Orthopedic}) - 0.0727(\text{Radiology}) + 0.3578(\text{Unknown}) \\
&+ 0.0451(\text{Other/Rare Specialty}) \\
&- 0.7437(\text{Nonconforming Material/Component}) \\
&- 0.2472(\text{Process Change Control}) - 0.8670(\text{Process Control}) \\
&- 0.2774(\text{Software Design}) - 0.4653(\text{Under Investigation by Firm}) \\
&- 0.7607(\text{Unknown/Undetermined by Firm}) - 0.2903(\text{Use Error}) \\
&- 0.5295(\text{Other Root Cause}) \\
&- 0.01584(\text{Products in Event}) - 0.40694(\text{K Numbers}) \\
&+ 0.000078(\text{Reason Length}).
\end{aligned} \tag{4}$$

$$\sigma = 0.466 \tag{5}$$

$$\text{Time Ratio} = e^{\beta_j} \tag{6}$$

The general AFT model is shown in Equation 3, while the fitted log-logistic model is shown in Equation 4. Time ratios were obtained using Equation 6.

Unlike the standard Cox proportional hazards model, the AFT framework does not require constant hazard ratios over time and directly models the relationship between predictor variables and survival duration. However, the selection of the the log - logistic distribution base on AIC does not independently establish model adequacy. Therefore, additional model diagnostics and predictive validation were necessary to assess

the suitability of the selected distribution and its underlying assumptions.

3.2 Log Logistic AFT Model and Assumptions

Scatterplots were constructed to examine the relationships between three numerical predictors: "number of products involved in the recall", "number of associated 510(k) numbers, and "recall reason description length" against their logarithmic follow - up time. The plots revealed substantial variability in log follow - up time across all three predictors. Both the "number of products involved" and " the number of associated 510(k) numbers" exhibited highly concentrated distributions at lower values, with relative few observations at higher counts. "Recall reason description length" demonstrated a wider range of observed values, although no clear line relationship with log follow - up time was apparent. These findings suggest that the relationships between the numerical predictors and recall duration may not be adequately characterized by simple linear functions. Consequently, natural cubic splines were considered within the log - logistic accelerated failure time framework to allow for nonlinear predictor effects. (Figure 6).

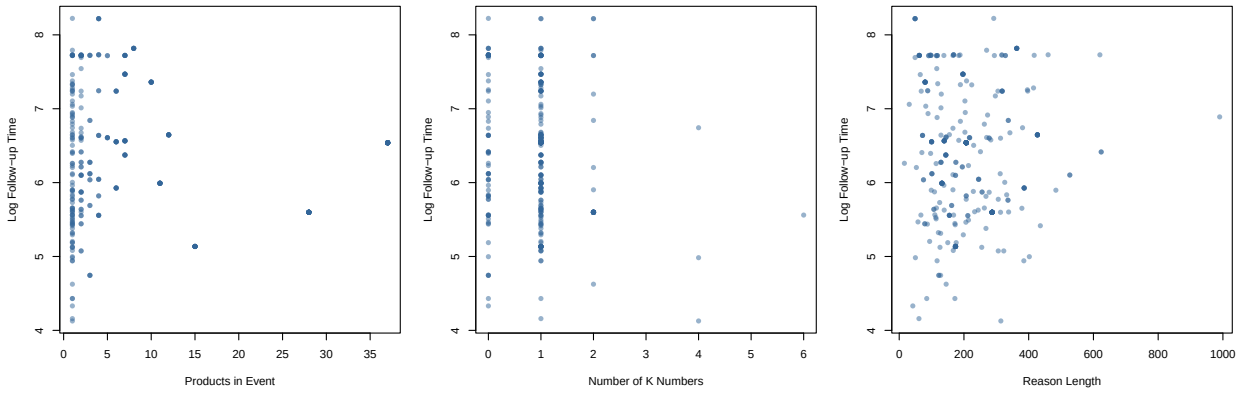


Figure 6: Scatter Plots of Log Follow-up Duration Against Continuous Covariates: Products in Event, Number of K Numbers, and Reason Length (1×3 Diagnostic Horizon Layout).

3.3 Non Linear Extension: Natural Cubic Spline AFT Framework

A log - logistic accelerated failure time model incorporating natural cubic splines was fitted to investigate potential nonlinear relationships between numerical predictors and medical device recall resolution times. Natural cubic spline functions with three degrees of freedom were applied to the "number of products involved in each recall", "the number of associated 510(k) numbers", and "recall reason description length". "Device classification", "medical specialty", and "root cause description" were retained as categorical predictors.

$$f_1(x) = \gamma_{11}B_{11}(x) + \gamma_{12}B_{12}(x) + \gamma_{13}B_{13}(x) \tag{7}$$

$$\begin{aligned} f_1(x) &= \sum_{k=1}^{3} \gamma_{1k}B_{1k}(x) \\ f_2(x) &= \sum_{k=1}^{3} \gamma_{2k}B_{2k}(x) \\ f_3(x) &= \sum_{k=1}^{3} \gamma_{3k}B_{3k}(x) \end{aligned} \tag{8}$$

where $f_1(x)$ is the spline function for " products in event", $f_2(x)$ is the spline function for "number of 510(k) numbers", $f_3(x)$ is the spline function for "recall reason description length", γ_{jk} is the estimated regression coefficient for spline basis k and predictor j, $B_{jk}(x)$ is the natural cubic spline basis function.

The overall likelihood ratio test indicated that the spline enhanced model was statistically significant ($\chi^2 = 164.3, df = 30, p-value < 0.001$) compared against the an intercept only model. Further, a likelihood test comparing the original linear log - logistic AFT model with the spline enhanced log - logistic AFT model produced a chi - square statistic of $\chi^2 = 64.454$ with six degrees of freedom ($p-value < 0.001$)), indicating a statistically significant improvement in model fit following the inclusion of nonlinear predictor functions.

Further, the spline enhanced AFT model results in an AIC reduction of 52.454, indicating a substantial improvement in relative model fit after accounting for the additional model complexity. These findings provide evidence that allowing nonlinear relationships between numerical predictors and log recall resolution time improves the model's ability to describe the observed training data. Consequently, the spline enhanced log logistic ATF model is preferred based on AIC, subject to further diagnostic assessment and validation on the independent testing dataset.

Table 8: Comparison of Linear and Spline-Enhanced Log-Logistic AFT Models

Model	**Parameters**	-2ℓ	**AIC**
Linear Log-logistic AFT	26	4348.023	4400.023
Spline Log-logistic AFT	32	4283.569	4347.569

Several categorical predictors demonstrated statistically significant associations with recall resolution times. Class II devices were associated with longer predicted resolution durations compared with Class I devices ($TR = 2.385, p-value < 0.001$). Similarly, "General/Plastic Surgery" devices exhibited longer predicted resolution times relative to Anesthesiology ($TR = 2.097, p-value = 0.017$). In contrast, Orthopedic devices ($TR = 0.509, p-value < 0.031$) and several "recall root cause" categories were associated with

shorter predicted resolution times relative to their corresponding reference groups.

The inclusion of natural cubic splines allowed the model to accommodate nonlinear associations between numerical predictors and log resolution time. Since spline coefficients represent components of basis functions, their individual estimates are not interpreted as direct time ratios. Instead, the combined spline functions are evaluated to characterize predictor effects across their observed ranges. Overall, the results suggest that incorporating nonlinear numerical predictor effects improves the fit of the log - logistic AFT model, supporting further examination of the spline enhanced model through likelihood ratio testing, diagnostic assessment, and out of sample validation.

$$\log(T_i) = \beta_0 + \mathbf{X}_i^\top \boldsymbol{\beta} + f_1(\text{products_in_event}_i) + f_2(\text{n_k_numbers}_i) + f_3(\text{reason_length_chars}_i) + \sigma\varepsilon_i, \quad (9)$$

$$\begin{aligned}
\log(T_i) = {} & 7.4739 + 0.8692(\text{Device Class 2}) - 0.3466(\text{Other/Rare Device Class}) \\
& - 0.4246(\text{Cardiovascular}) + 0.1315(\text{Clinical Chemistry}) + 0.0032(\text{Dental}) \\
& - 0.4924(\text{Gastroenterology/Urology}) - 0.0727(\text{General Hospital}) \\
& + 0.7407(\text{General/Plastic Surgery}) - 0.5586(\text{Neurology}) \\
& - 0.6754(\text{Orthopedic}) - 0.0017(\text{Radiology}) + 0.0202(\text{Unknown}) \\
& + 0.0886(\text{Other/Rare Specialty}) \\
& - 0.6529(\text{Nonconforming Material/Component}) \\
& - 0.6633(\text{Process Change Control}) - 0.6313(\text{Process Control}) \\
& - 0.3023(\text{Software Design}) - 0.3598(\text{Under Investigation by Firm}) \\
& - 0.6578(\text{Unknown/Undetermined by Firm}) + 0.0555(\text{Use Error}) \\
& - 0.4563(\text{Other Root Cause}) \\
& + f_1(\text{Products in Event}) + f_2(\text{K Numbers}) + f_3(\text{Reason Length}),
\end{aligned} \quad (10)$$

$$\begin{aligned}
f_1(x) &= -1.1076\, B_{11}(x) + 0.5179\, B_{12}(x) - 0.3693\, B_{13}(x), \\
f_2(x) &= 0.6237\, B_{21}(x) - 1.5602\, B_{22}(x) - 1.3273\, B_{23}(x), \\
f_3(x) &= 0.2709\, B_{31}(x) - 1.8214\, B_{32}(x) - 0.2342\, B_{33}(x),
\end{aligned} \quad (11)$$

ns(variable, df = 3)

$$\sigma = 0.416.$$

$$\chi^2(30) = 164.3, \qquad p = 1.8 \times 10^{-20}.$$

$$\chi^2(6) = 64.45, \qquad p = 5.58 \times 10^{-12}.$$

$$\mathrm{AIC}_{\mathrm{linear}} = 4400.02, \qquad \mathrm{AIC}_{\mathrm{spline}} = 4347.57.$$

	Term	Coefficient	Std_Error	z	p_value
(Intercept)	(Intercept)	7.4739	0.4242	17.6173	0.0000
device_class2	device_class2	0.8692	0.1953	4.4515	0.0000
device_classOther/Rare	device_classOther/Rare	-0.3466	0.6633	-0.5225	0.6013
medical_specialtyCardiovascular	medical_specialtyCardiovascular	-0.4246	0.3305	-1.2845	0.1990
medical_specialtyClinical Chemistry	medical_specialtyClinical Chemistry	0.1315	0.3465	0.3796	0.7043
medical_specialtyDental	medical_specialtyDental	0.0032	0.4786	0.0067	0.9947
medical_specialtyGastroenterology, Urology	medical_specialtyGastroenterology, Urology	-0.4923	0.3909	-1.2595	0.2079
medical_specialtyGeneral Hospital	medical_specialtyGeneral Hospital	-0.0727	0.2962	-0.2453	0.8062
medical_specialtyGeneral, Plastic Surgery	medical_specialtyGeneral, Plastic Surgery	0.7406	0.3112	2.3801	0.0173
medical_specialtyNeurology	medical_specialtyNeurology	-0.5586	0.3408	-1.6389	0.1012
medical_specialtyOrthopedic	medical_specialtyOrthopedic	-0.6754	0.3127	-2.1603	0.0308
medical_specialtyRadiology	medical_specialtyRadiology	-0.0017	0.3243	-0.0053	0.9957
medical_specialtyUnknown	medical_specialtyUnknown	0.0202	0.7481	0.0270	0.9785
medical_specialtyOther/Rare	medical_specialtyOther/Rare	0.0886	0.3095	0.2864	0.7746
root_cause_descriptionNonconforming Material/Component	root_cause_descriptionNonconforming Material/Component	-0.6529	0.1770	-3.6896	0.0002
root_cause_descriptionProcess change control	root_cause_descriptionProcess change control	-0.6633	0.3199	-2.0733	0.0381
root_cause_descriptionProcess control	root_cause_descriptionProcess control	-0.6313	0.1866	-3.3831	0.0007
root_cause_descriptionSoftware design	root_cause_descriptionSoftware design	-0.3023	0.3134	-0.9648	0.3347
root_cause_descriptionUnder Investigation by firm	root_cause_descriptionUnder Investigation by firm	-0.3598	0.1836	-1.9600	0.0500
root_cause_descriptionUnknown/Undetermined by firm	root_cause_descriptionUnknown/Undetermined by firm	-0.6578	0.2657	-2.4757	0.0133
root_cause_descriptionUse error	root_cause_descriptionUse error	0.0555	0.3007	0.1844	0.8537
root_cause_descriptionOther	root_cause_descriptionOther	-0.4563	0.1830	-2.4939	0.0126
ns(products_in_event, df = 3)1	ns(products_in_event, df = 3)1	-1.1076	0.3698	-2.9948	0.0027
ns(products_in_event, df = 3)2	ns(products_in_event, df = 3)2	0.5179	0.2456	2.1083	0.0350
ns(products_in_event, df = 3)3	ns(products_in_event, df = 3)3	-0.3693	0.1961	-1.8837	0.0596
ns(n_k_numbers, df = 3)1	ns(n_k_numbers, df = 3)1	0.6237	0.6800	0.9173	0.3590
ns(n_k_numbers, df = 3)2	ns(n_k_numbers, df = 3)2	-1.5602	0.4700	-3.3198	0.0009
ns(n_k_numbers, df = 3)3	ns(n_k_numbers, df = 3)3	-1.3273	0.6863	-1.9338	0.0531
ns(reason_length_chars, df = 3)1	ns(reason_length_chars, df = 3)1	0.2709	0.2787	0.9720	0.3311
ns(reason_length_chars, df = 3)2	ns(reason_length_chars, df = 3)2	-1.8214	0.6773	-2.6894	0.0072
ns(reason_length_chars, df = 3)3	ns(reason_length_chars, df = 3)3	-0.2342	0.6265	-0.3737	0.7086
Log(scale)	Log(scale)	-0.8771	0.0514	-17.0756	0.0000

3.4 Error Diagnosis

Response residuals were examined to assess the fit of the spline enhanced log -logistic AFT model and identify potentially unusual observations. The residual plot displayed observations on both sides of the reference line with a substantial of residuals surrounding zero. No obvious systematic trend was observed across the observation indices, although several observations exhibited relatively large residuals suggesting that the fitted model does not adequately predict the observed follow up durations for certain observations.

Although the residual plot provides a preliminary assessment of model behavior, the presence of censored observations limits the interpretation of conventional response residuals in survival analysis. Further, the absence of a visible pattern across the observations does not independently establish that the assumed log - logistic distribution or the spline functional forms are appropriate. Hence, additional survival specific diagnostic methods, including the Cox - Snell residual analysis were considered to further evaluate model adequacy. (7).

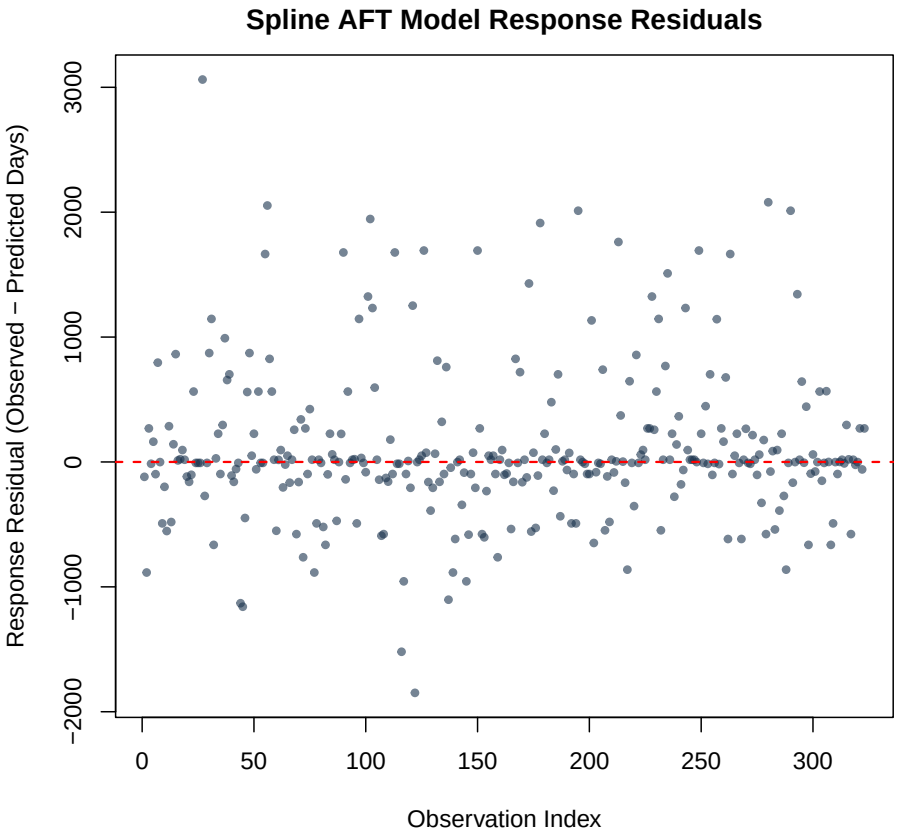


Figure 7: Scatter Plot of Spline Log-Logistic AFT Model Response Residuals Around the Zero-Variance Baseline Configuration ($e_i = Y_i - \hat{Y}_i$).

3.5 Goodness-of-Fit Validation: Cox-Snell Residual Tracking

A Cox - Snell residual analysis was conducted to assess the goodness of fit of the spline enhanced log logistic AFT model. The Cox -Snell residuals were calculated using the estimated conditional survival function, and their cumulative hazard was compared with the theoretical unit exponential reference line. Under adequate model specification, the estimated cumulative hazard of the Cox - Snell residuals is expected to approximately follow a straight line passing through the origin.

The diagnostic plot demonstrated relatively close agreement with the theoretical reference line at the smaller residual values, implying a reasonable fit over the early portion of the residual distribution. However, deviations become larger as the residual values increased, with the estimated cumulative hazard falling below the theoretical reference line and eventually approaching a plateau. This departure suggests potential

limitations in the model's ability to characterize the upper tail of the survival time distribution. Hence, the interpretation of these deviations required caution since fewer observations remain at risk at larger residual values, increasing uncertainty in the cumulative hazard estimates.

Overall, the Cox -Snell residual diagnostic suggests that the spline enhanced log - logistic AFT model provides reasonable approximation over part of the observed survival time distribution, though deviations at larger residual values indicate that the model may not fully capture the tail behavior.

(Figure 8).

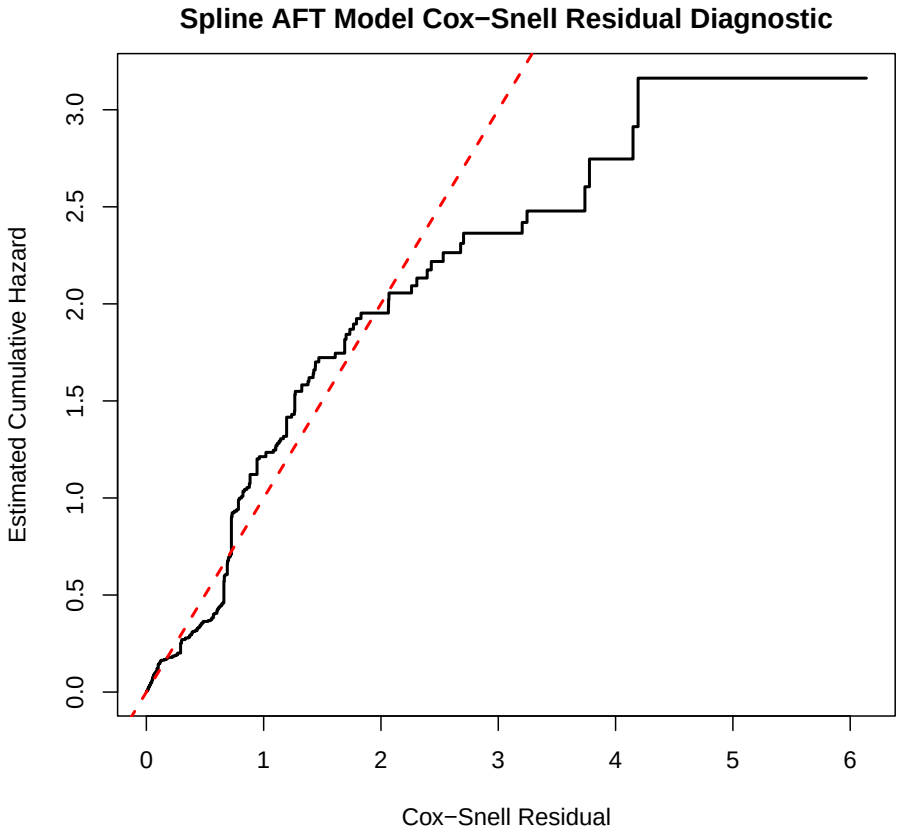


Figure 8: Cox-Snell Residual Cumulative Hazard Trajectory Evaluation

3.6 Multicollinearity

Finally, multicollinearity among the predictor variables was assessed using generalized variance inflation factors (GVIF). An auxiliary linear regression model was fitted using the logarithm of follow - up time as the response variable and the same predictors included in the original log - logistic AFT model. Since

Table 9: Generalized Variance Inflation Factors

Predictor	GVIF	DF	Adjusted_GVIF
device_class	22.248	2	2.172
medical_specialty	154.996	11	1.258
root_cause_description	11.652	8	1.166
products_in_event	2.341	1	1.530
n_k_numbers	2.155	1	1.468
reason_length_chars	1.190	1	1.091

several categorical predictors contained multiple levels, adjusted GVIF values were examined to account for differences in degrees of freedom.

$$GVIF_{\text{adjusted}} = GVIF^{\frac{1}{2df}} \tag{12}$$

Device classification exhibited the highest adjusted GVIF (2.172), followed by the number of products involved in each recall (1.530), the number associated 510(k) numbers (1.468), medical specialty (1.258), root cause description (1.166), and recall reason description length (1.091). Overall, the adjusted GVIF values did not indicate substantial multicollinearity among the examined predictors, suggesting that predictor redundancy was unlikely to be a major source of instability in the original linear effects model.

4 CONCLUSION AND FUTURE WORK

This study investigated factors associated with medical device recall resolution times using survival analysis techniques, with particular emphasis on comparing the Cox proportional hazards model and the parametric accelerated failure time (AFT) models. The analysis included 404 medical device recall observations and examined the relationships between recall resolution duration and several device characteristics including device classification, medical specialty, root cause description, number of products involved, number of associated 510(k) numbers, and recall reason description length.

Initial exploratory data analysis revealed substantial variation in recall characteristics, with Class II devices representing the majority of observations. Kaplan - Meier survival analysis provided a nonparametric assessment of recall resolution patterns, while the Cox proportional hazards model was initially employed to identify factors associated with resolution hazards. Although the Cox model demonstrated overall statistical significance, Schoenfeld residual diagnostics indicated a substantial violation of the proportional hazards assumption (global p value ¡ 0.001). Consequently, alternative survival modeling approaches were investigated.

Four parametric AFT models were evaluated using exponential, Weibull, lognormal, and log - logistic distributions. Based on the Akaike Information Criterion (AIC), the log - logistic AFT model provided the most favorable balance between goodness of fit and model complexity among the candidate distributions. To further investigate potentially nonlinear relationships between numerical predictors and recall resolution times, natural cubic splines were incorporated into the log logistic AFT framework.

The spline enhanced log - logistic AFT model demonstrated a substantial improvement over the original linear effects model, reducing the AIC from 4400.02 to 4347.57. A likelihood ratio test confirmed that the additional spline terms collectively improved model fit ($\chi^2 = 64.45, df = 6, p-value < 0.001$). These findings imply that allowing flexible relationships between numerical predictors and log resolution time provides a more suitable representation of the observed training data than the original linear model.

Several categorical predictors demonstrated statistically significant association with recall resolution duration. In particular, Class II devices were associated with longer predicted resolution times compared with Class I devices, while certain medical specialties and recall root causes were associated with shorter or longer predicted durations. These findings highlight the potential importance of device characteristics and recall circumstances in understanding variation in medical device recall resolution times.

Model diagnostics provided additional insight into the adequacy of the spline enhanced log logistic AFT model. Response residual analysis identified some observations with relatively large deviations from fitted

values, while Cox - Snell residual diagnostics revealed departures from the theoretical reference distribution, particular at larger residual values. These findings suggest that the spline enhanced log logistic (AFT) model may not fully capture the upper tail of the recall resolution time distribution. Further, the preliminary multicollinearity assessment did not identify substantial predictor redundancy.

Overall, the results support the spline enhanced log - logistic AFT model as the preferred model among those evaluated based on in sample goodnes of fit. The study demonstrates the importance of assessing model assumptions, comparing alternative survival distributions, and considering nonlinear predictor effects when analyzing medical device recall resolution times.

Future potential work would focus on evaluating and improving the predictive performance, robustness, and practical applicability of the spline enhanced log - logistic AFT model. An important next step is to assess model performance using the independent testing dataset consisting of 81 observations. Predictive performance could be evaluated using survival specific measures such as the concordance index, integrated Brier score, and time - dependent prediction error. These assessments would help determine whether the improved goodness of fit observed in the training dataset translates into better predictive performance on previously unseen observations.

5 REFERENCES

References

[1] Collett, David. Modelling Survival Data in Medical Research. 4th ed., Chapman and Hall/CRC, 2023

[2] Feng, Kyle. "Medical Device Recall Survival Data." Kaggle, https://www.kaggle.com/datasets/kylefengkfeng209/medical device-recall-survival-data. Accessed 28 Oct. 2026.

[3] Korosteleva, Olga. Nonparametric Methods in Statistics with SAS Applications. Chapman & Hall/CRC, 2013

[4] Klein, John P., and Melvin L. Moeschberger. Survival Analysis: Techniques for Censored and Truncated Data. 2nd ed., Springer, 2003

[5] Wackerly, Dennis, et al. Mathematical Statistics with Applications. 8th ed., Cengage Learning, 2026.